Magnetic Field from a Force Balance, and Dipole Energy

Goal.

- Apply the velocity-selector force balance to solve for the *field* instead of the speed — same equation, different unknown.
- Use $U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos\theta$ to find a dipole's potential energy at a given orientation, and the *change* in energy between two different orientations.

1. Worked example: finding the field in a velocity selector

Recall the velocity selector: a charge moving through crossed $\vec{E}$ and $\vec{B}$ fields goes straight only when the electric and magnetic forces exactly cancel, $qE = qvB$. Last time you solved that for the speed v. This time you're given the speed and have to solve for the field.

A charged particle moves with speed $v = 3.0 \times 10^5$ m/s through a region containing a uniform electric field $E = 6.0 \times 10^4$ N/C, directed vertically downward. The particle passes straight through without deflecting. What is the magnitude of the magnetic field $\vec{B}$ present in the region? Assume $\vec{B}$ is perpendicular to both $\vec{v}$ and $\vec{E}$, and the particle is positively charged.

MODEL. The particle goes straight only if the electric force and the magnetic force on it are _____, exactly canceling.

SOLVE. Set the force magnitudes equal:

$$qE = _____$$

The charge q _____ — the balance condition doesn't depend on it. Solve for B:

$B =$

Plug in the numbers:

$B =$

Direction. The electric force on the positive charge points downward (along $\vec{E}$), so the magnetic force $q\vec{v} \times \vec{B}$ must point _____ to cancel it. With $\vec{v}$ horizontal, the right-hand rule then fixes which way $\vec{B}$ must point.

Check. If the particle's speed were doubled but E stayed the same, would the required B go up, go down, or stay the same? Does that match your intuition — does a faster particle need a *stronger* or *weaker* field to still be balanced?

2. Worked example: energy of a magnetic dipole, two orientations

Recall the energy of a magnetic dipole in a field: $U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos\theta$, lowest when $\vec{\mu}$ is aligned with $\vec{B}$ ($\theta = 0$) and highest when anti-aligned ($\theta = 180°$).

A magnetic dipole with moment $\mu = 0.60$ $A{\cdot}m^2$ sits in a uniform field $B = 0.30$ T. Someone rotates it from $\theta_1 = 40°$ to $\theta_2 = 110°$ (both measured from $\vec{B}$). What is the change *in the dipole's potential energy?*

MODEL. The energy change is just the energy in the final orientation minus the energy in the initial one: $\Delta U =$ _____.

SOLVE. Find the energy at each angle separately:

$$U_1 = -\mu B \cos\theta_1 = _____$$

$$U_2 = -\mu B \cos\theta_2 = _____$$

Now subtract:

$\Delta U = U_2 - U_1 =$

Plug in the numbers:

$\Delta U =$

Check. Is ΔU positive or negative here? $\theta_1 = 40°$ is closer to alignment ($0°$) than $\theta_2 = 110°$ is. Does the sign of ΔU match the fact that torque always twists a dipole *toward* alignment — so moving *away* from alignment should take energy *in*, not give it up?